基于MATLAB的智能驾驶定位开发与高精地图应用



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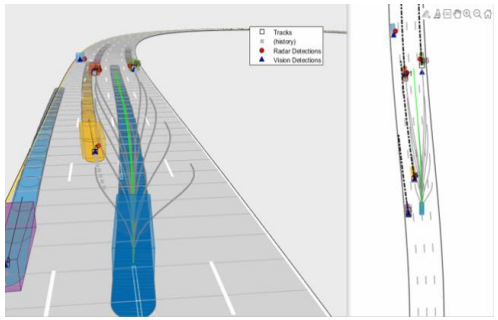
2021-11-30

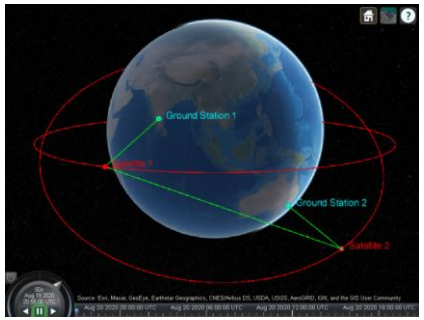
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内容简介

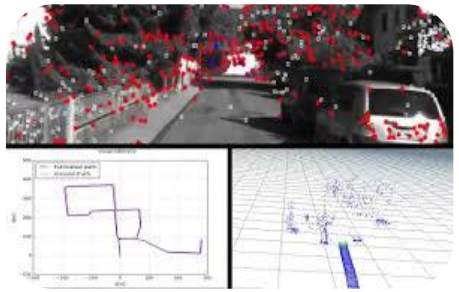
- 基于MATLAB的智能驾驶定位开发 (MathWorks)
  - 基于高精地图数据的高速场景定位
  - 基于仿真的泊车场景视觉定位
- 高精地图应用 (EMG)
  - 高精地图数据服务
  - 基于RoadRunner及高精地图数据的仿真

定位技术及智能驾驶应用场景

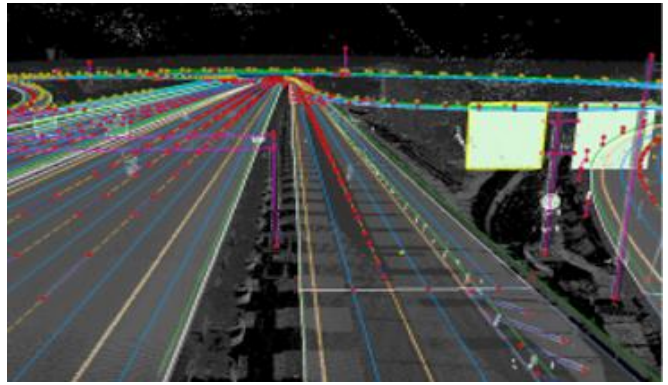


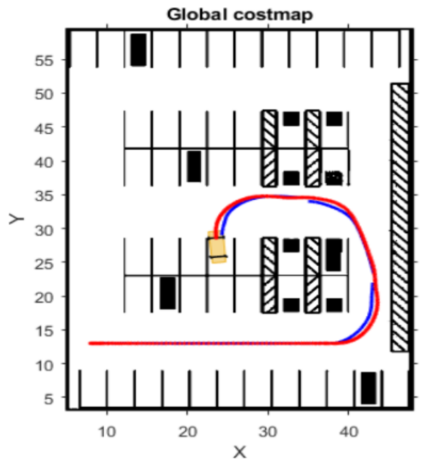


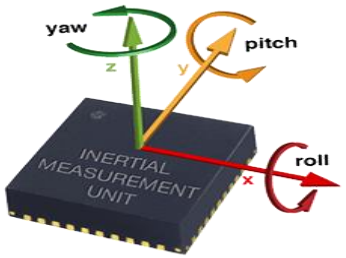
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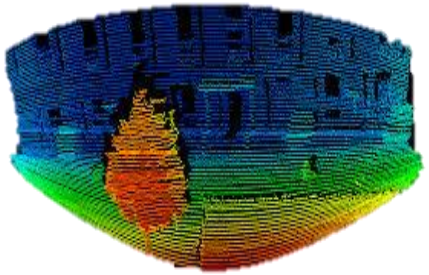
Visual



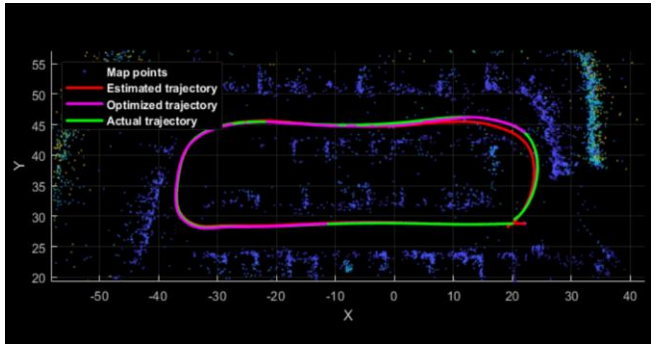




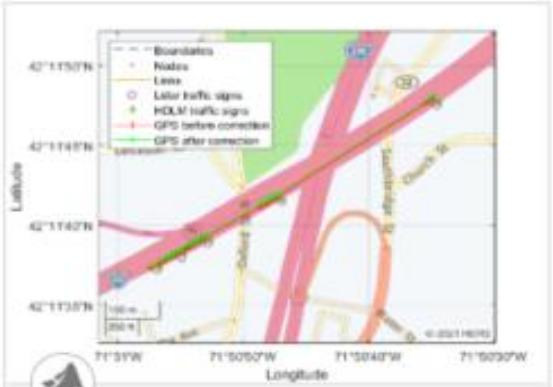
IMU



LIDAR



基于MATLAB的定位示例

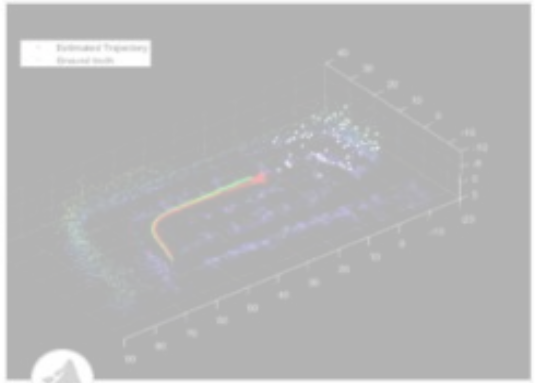


Localization Correction Using Traffic Sign Data from HERE HD Maps

Use HERE HD Live Map (HDLM) traffic sign data to correct GPS measurements from an autonomous vehicle.

Open Live Script

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Visual Localization in a Parking Lot

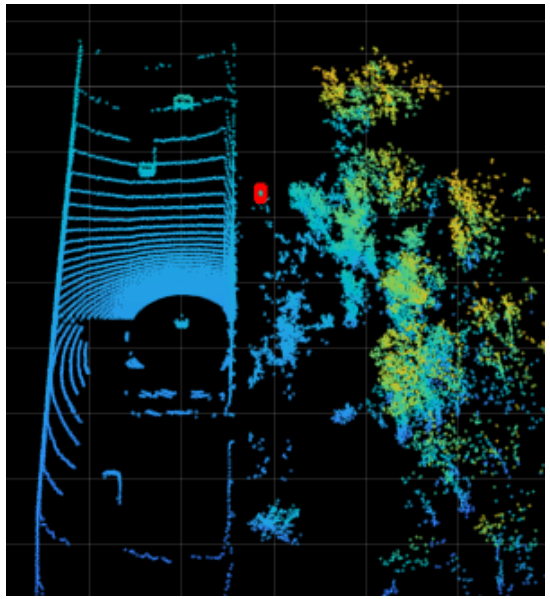
Develop a visual localization system using synthetic image data from a parking lot scene in Unreal Engine environment.

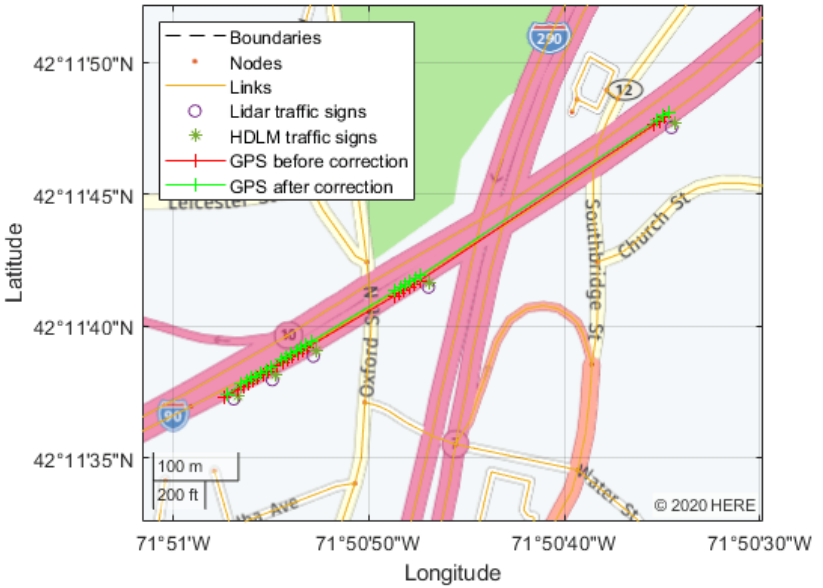
Open Live Script

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示例一：高速场景基于高精地图的定位







- Preprocess and sync data from ROS
- Read traffic sign information from HERE HD Live Map
- Match signs detected by the onboard sensors with signs stored map data
- Improve localization accuracy by combining GPS measurements with map data

[Localization Correction Using Traffic Sign Data from HD Maps]

Automated Driving Toolbox™

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获取HERE HDLM高精地图数据

GPS Data

Tiles



Layers

Road Centerline Model

- TopologyGeometry
- RoutingAttributes

HD Lane Model

- LaneTopology
- LaneGeometryPolyline

Localization Model

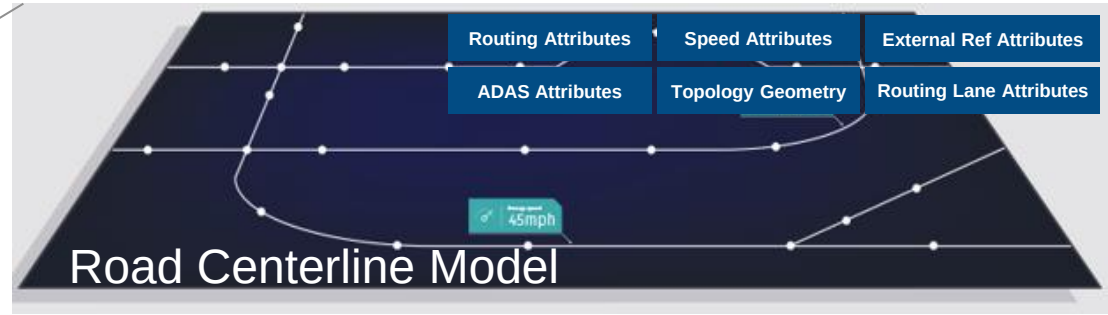
- LocalizationBarrier
- LocalizationSign

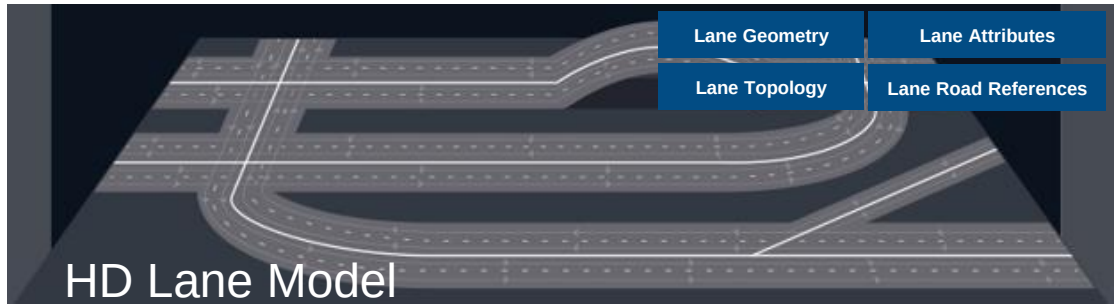
Variables - reader.Layers

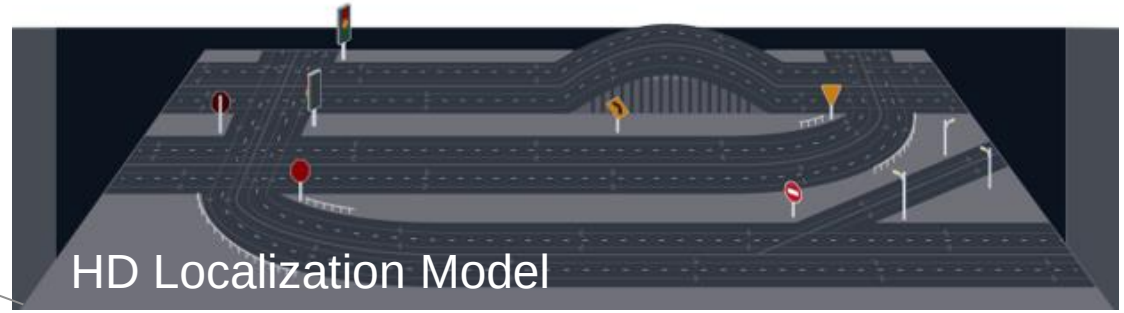
+8 | reader | reader.Layers

reader.Layers

	1	2
1	AdasAttributes	
2	ExternalReference...	
3	LaneAttributes	
4	LaneGeometryPoly...	
5	LaneRoadReferenc...	
6	LaneTopology	
7	LocalizationBarrier	
8	LocalizationPole	
9	LocalizationSign	
10	RoutingAttributes	
11	RoutingLaneAttrib...	
12	SpeedAttributes	
13	TopologyGeometry	
14		



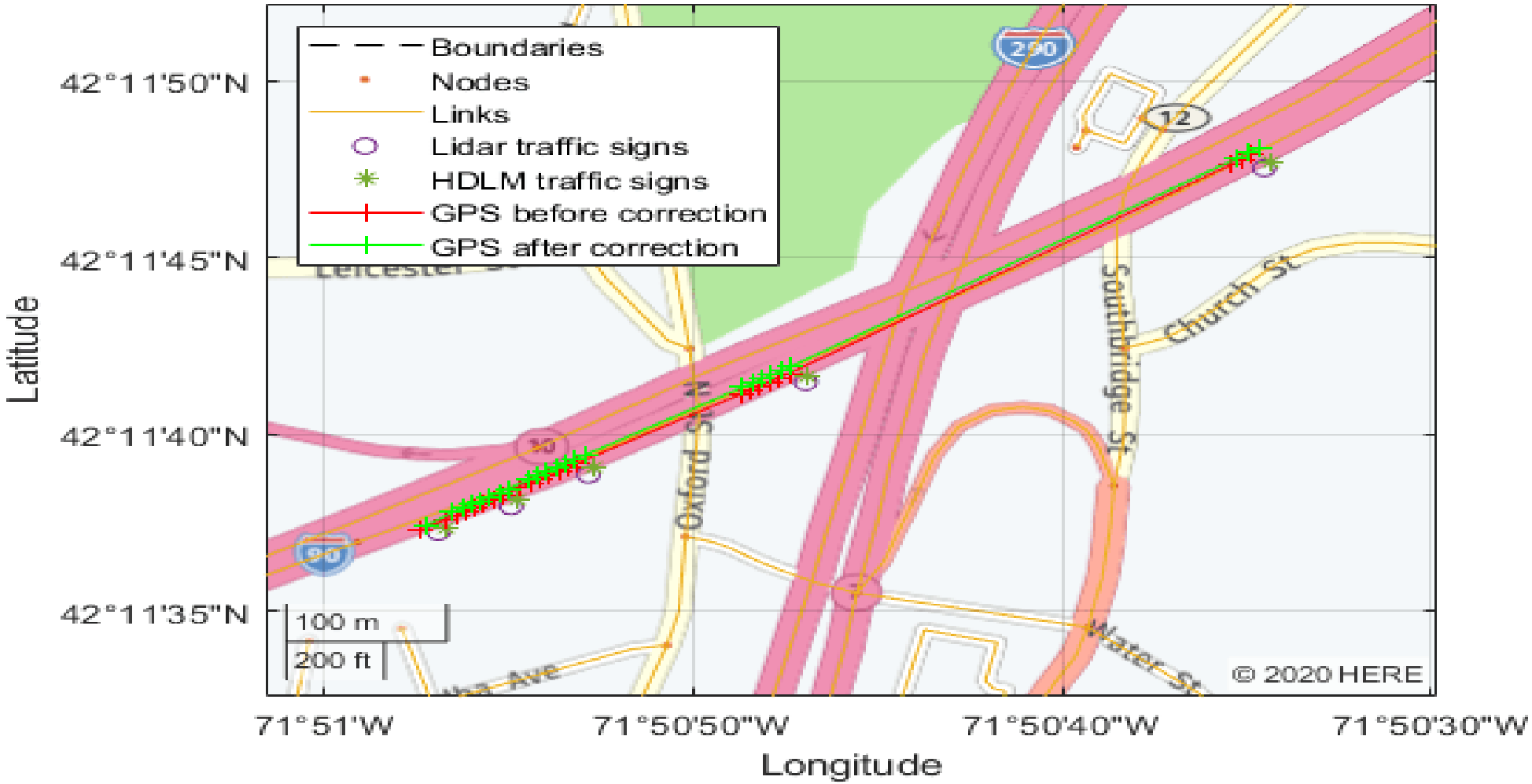





```
config = hereHDLMConfiguration('hrn:here:data::olp-here-had:here-hdlm-protobuf-na-2');
reader = hereHDLMReader(gpsRoute(:,1),gpsRoute(:,2),'Configuration',config);

topologyLayer = read(reader,'TopologyGeometry');
signLayer= read(reader,'LocalizationSign');
```

高精地图辅助定位结果



[Localization Correction Using Traffic Sign Data from HD Maps]

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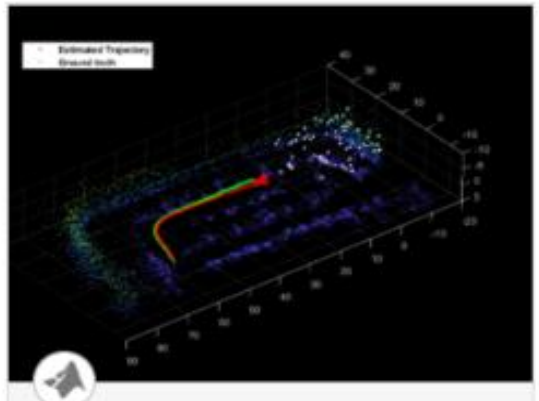
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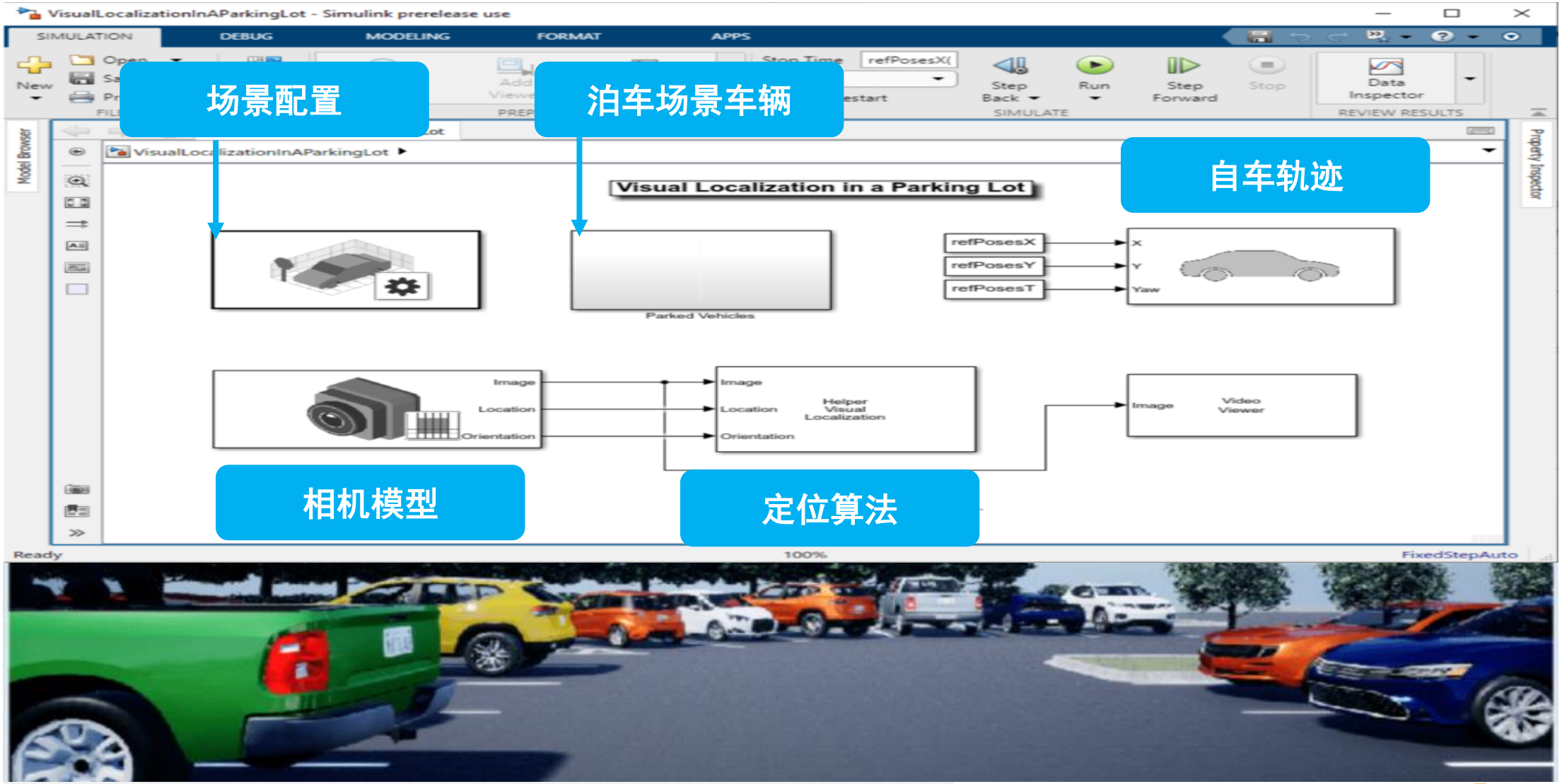
Visual Localization in a Parking Lot

Develop a visual localization system using synthetic image data from a parking lot scene in Unreal Engine environment.

Open Live Script

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示例二：基于仿真的泊车场景视觉定位

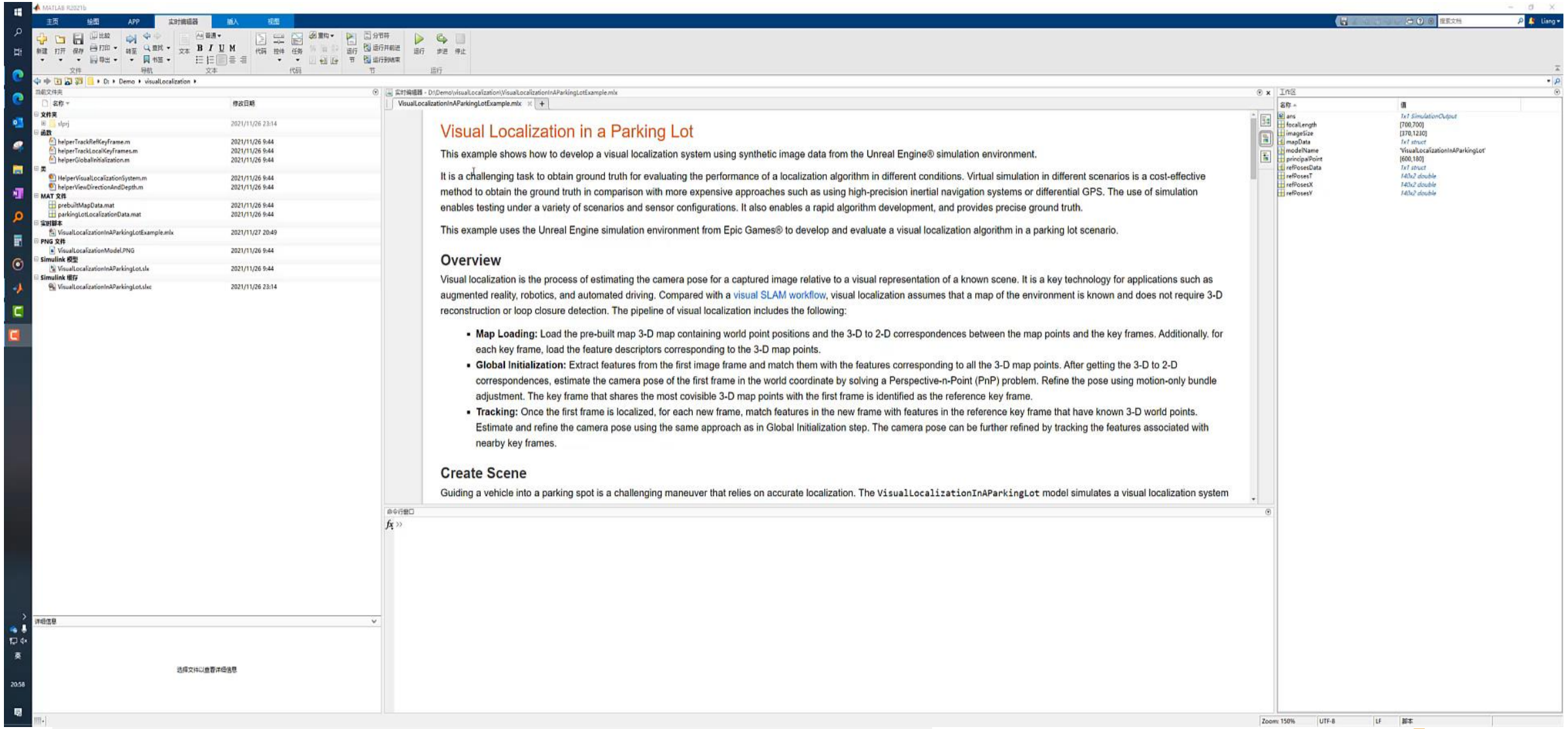


[Visual Localization in a Parking Lot](#)

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泊车场景基于视觉的定位

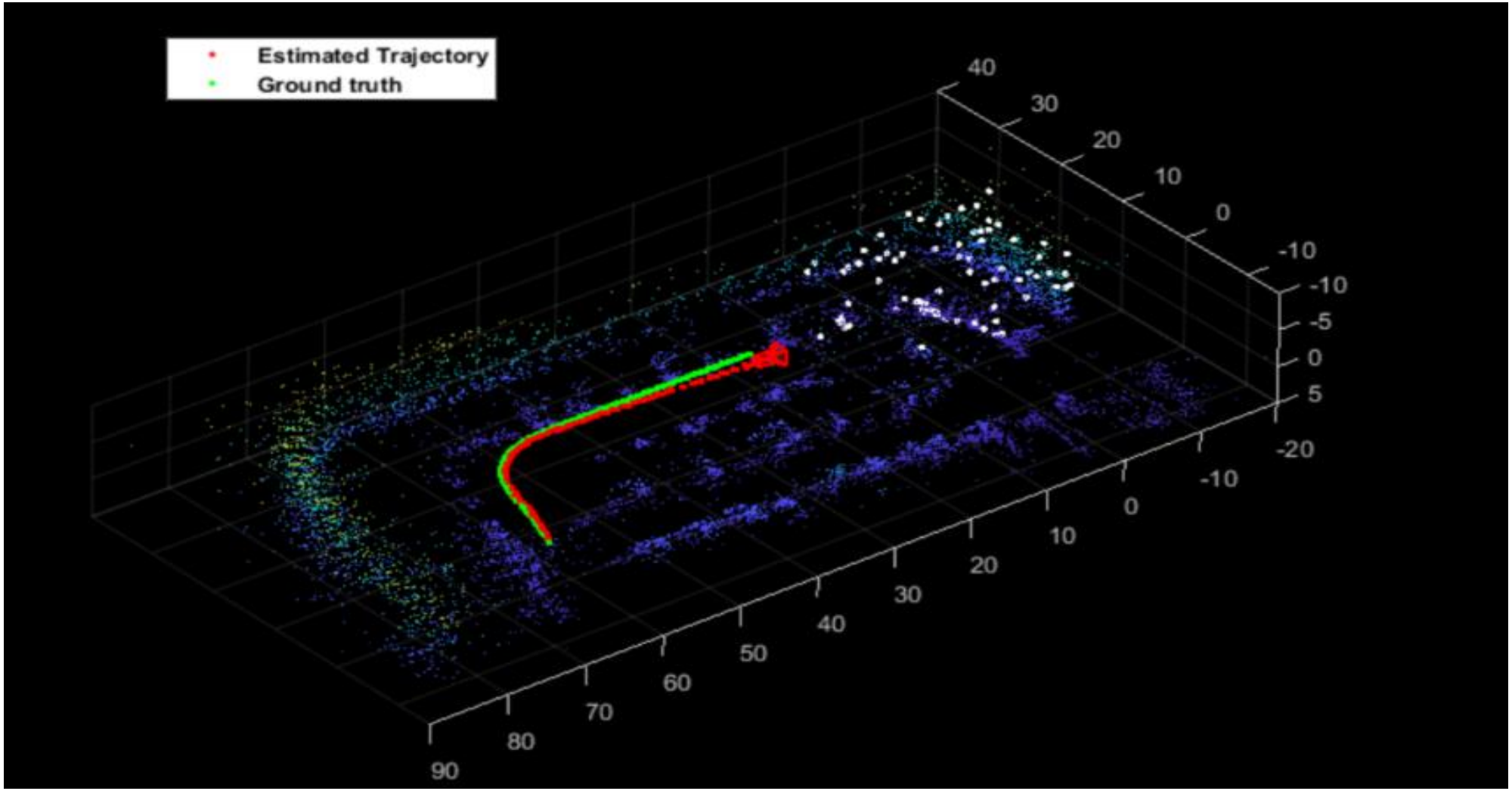


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泊车场景基于视觉的定位



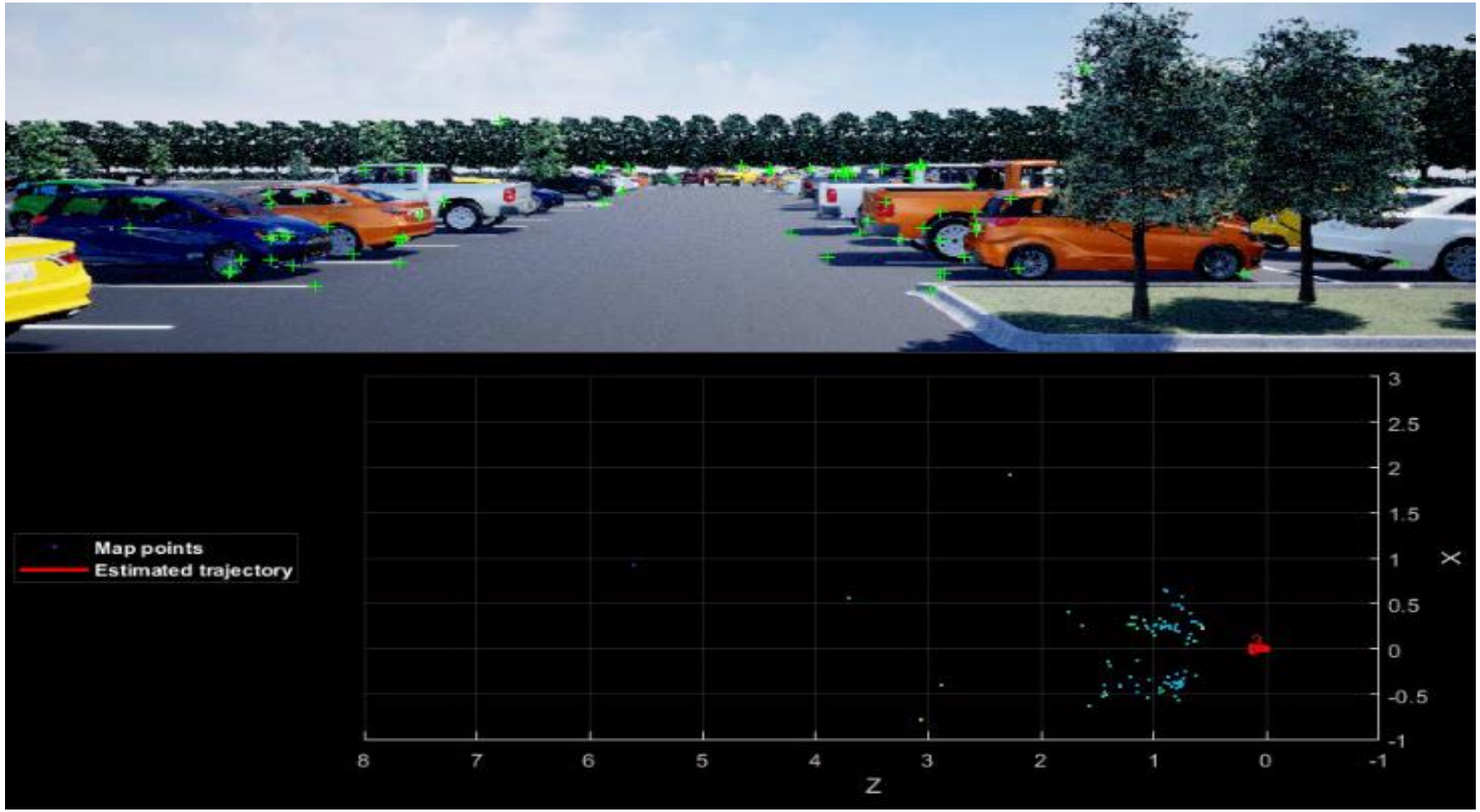
[Visual Localization in a Parking Lot]

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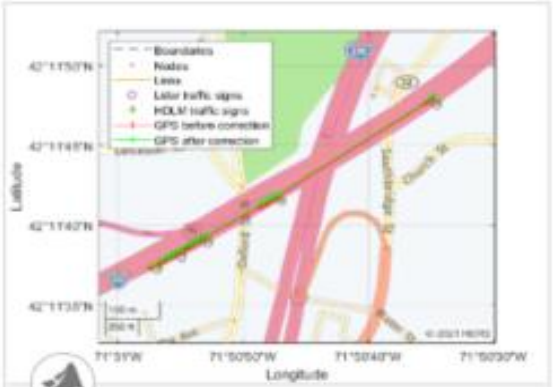
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泊车场景下的Visual-SLAM

Scene Configuration → Camera Configuration → 3-D Simulation → Algorithm Testing



基于MATLAB的定位示例

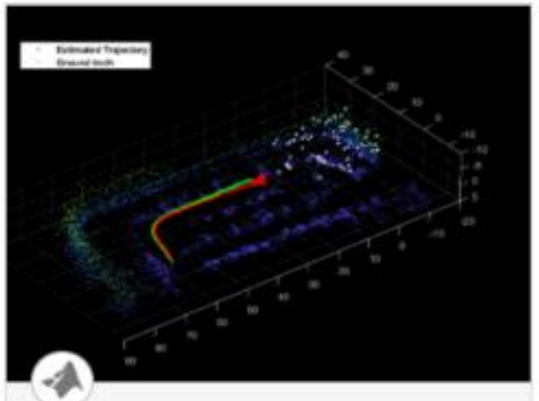


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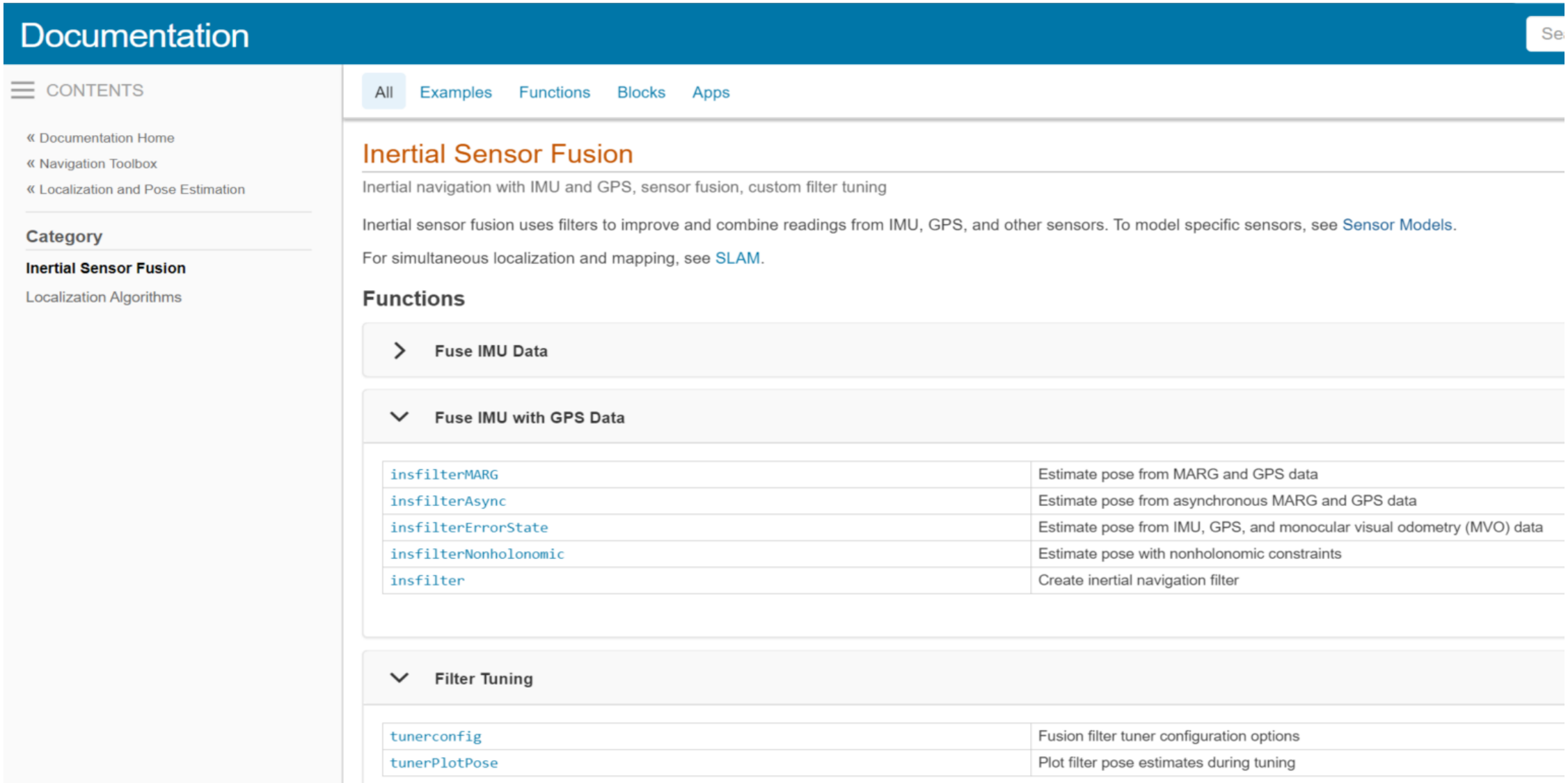
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IMU与GPS融合



Documentation

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Category

Inertial Sensor Fusion

Localization Algorithms

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Inertial Sensor Fusion

Inertial navigation with IMU and GPS, sensor fusion, custom filter tuning

Inertial sensor fusion uses filters to improve and combine readings from IMU, GPS, and other sensors. To model specific sensors, see Sensor Models.

For simultaneous localization and mapping, see SLAM.

Functions

Fuse IMU Data

Fuse IMU with GPS Data

insfilterMARG	Estimate pose from MARG and GPS data
insfilterAsync	Estimate pose from asynchronous MARG and GPS data
insfilterErrorState	Estimate pose from IMU, GPS, and monocular visual odometry (MVO) data
insfilterNonholonomic	Estimate pose with nonholonomic constraints
insfilter	Create inertial navigation filter

Filter Tuning

tunerconfig	Fusion filter tuner configuration options
tunerPlotPose	Plot filter pose estimates during tuning

谢谢！